

RESUMPTION

Helix OTA — Session Resumption (canonical, always-current)

Field	Value
Revision	10
Last modified	2026-06-11T10:10:00Z
Status	active — the §11.4.131 single canonical out-of-the-box session-resumption file
Standard path	docs/RESUMPTION.md (this file — the fixed project-declared §11.4.131 entry point; do not move without a §11.4.66 operator decision) Point any fresh session at THIS file. It carries a SHORT one-line resume + a FULL block, the read-first handoff docs, exact live-state anchors, current PHASE + NEXT + terminal goal, and the binding constraints. Moment-valid for HEAD on main (all 4 remotes aligned). 2026-06-11: 🎉 PWU-AB-1 FULL A/B SLOT SWITCH PROVEN GREEN on this macOS host — real U-Boot 2024.01 + QEMU virt+HVF, same u-boot.bin+disk booted twice differing only in BOOT_ORDER routes root to a different physical slot, guest reports it from userspace (Run A→/etc/slot_id=A+/dev/vda2+helix_slot=A; Run B→B+/dev/vda3+helix_slot=B; both live shells; neither cross-boots). 10/10 anti-bluff asserts PASS, §11.4.50 determinism 3/3 identical , RED→GREEN §11.4.115, evidence docs/qa/20260611T094958Z-ab-slot-switch/ (240-line consoles + determinism_soak.txt). build8
Status summary	

Field	Value
	produced a genuine u-boot.bin (1.0M) + the 2-slot GPT ab_disk.img. NEXT: PWU-AB-2 RAUC dm-verity (ab_rauc_verity.sh authored — flags the real RAUC[?]boot.cmd env-scheme integration gap + needs a . raucb bundle), PWU-AB-3 corrupt-slot auto-rollback (ab_rollback.sh authored — bootcount>bootlimit BOOT_ORDER swap); both parse-clean + UNVERIFIED-pending-conductor-run. 2026-06-10→11 autonomous session: EVERY achievable software tier GREEN + §11.4.50 determinism soaks + sustained loadtest, THEN an operator-directed 6-stream parallel-hardening round (real coverage gains api 87.7%/fabric 96.7%/transport 94.9%, dashboard 58→93, §11.4.65 PDF backfill 66 siblings, extended security suite 26/0/0 ×3 — two real defects found+fixed honestly). Full §11.4.27 test-type matrix (unit/integration/e2e/full-automation/security/DDoS/scaling/chaos/stress/perf/UI/Challenges) re-confirmed. See docs/qa/STABILITY_REPORT.md Rev 2 + docs/qa/20260610T16{20Z-overnight-revalidation,40Z-determinism-soak}/. Android bricks product-identical to proof (pins 1061015/8bb8d2f unchanged), not re-run (§12.6/§11.4.6). BLOCKED (hardware): T2 Cuttlefish / GSI-A·B real-apply / T3 RK3588 → no release tag (§11.4.40 would be a bluff). Achievable+zero-risk queue is EXHAUSTED; remaining items are hardware-gated or correctly-deferred-as-risky (fabric scheduler = unwired §11.4.124).

SHORT — paste this first sentence into a fresh session

Read docs/research/main_specs/CONTINUATION.md first, run git fetch --all --prune, then continue the Helix OTA autonomous loop toward the next validated-and-published version tag — current PHASE is “OVERNIGHT STABLE — every achievable tier RE-VERIFIED GREEN on this HEAD (see docs/qa/STABILITY_REPORT.md Rev 2); P4 Cuttlefish

+ GSI-A/B + T3 RK3588 remain hardware-BLOCKED so no release tag yet (§11.4.40); achievable+zero-risk queue EXHAUSTED”; HEAD is 0d27438 on main (all 5 remotes aligned).

FULL — detailed resumption block

0. Read FIRST (in order)

1. docs/research/main_specs/CONTINUATION.md — the live work-state handoff (DONE / NEXT / locked decisions). **This is the primary handoff doc.**
2. .remember/remember.md — present but currently empty; no extra state there.
3. docs/design/EMULATED_DEVICE_TESTING.md — the tiered emulator-testing plan (Tier-1/2/3) now in flight.
4. Then run: `git fetch --all --prune` and reconcile HEAD. .@{u} before any edit (§11.4.37 fetch-before-edit).

1. Exact live-state anchors (moment-valid 2026-06-10)

- **HEAD commit:** a58f7f8 — test(parallel-round-2): integration-port fix + deviceemu/submodule coverage + e2e halt-safety + bank-wire (on main, all 5 remotes aligned). Recent chain: a58f7f8 (parallel round 2) ← fc383f2 (doc sync) ← dead6ef (PWU-5 security) ← a151bd2 (parallel round 1) ← overnight re-validation/soak ← eb9e1c4. **Build state: GREEN — comprehensively re-verified; see docs/qa/STABILITY_REPORT.md Rev 3.** Owned submodules advanced + pushed: ota-protocol → eda12b7 (→100% cov), ota-artifact-validator → 087fa08 (→100% cov).
- **Parallel-hardening round (operator-directed 6 subagent streams, all landed + pushed):** real anti-bluff coverage gains — internal/api 81.2→87.7%, internal/fabric 83.6→96.7%, internal/transport 79.5→94.9%, internal/rollout +11.3pts, internal/store +2.7pts (non-integration); **dashboard Vitest 58→93** (Audit/Deployments/Groups, ×3 deterministic); **§11.4.65 PDF backfill** (66 .html/.pdf siblings via new scripts/sync_md_siblings.sh); **extended security suite** tests/security/security_probes_extended.sh (6 probe families, **26/0/0 ×3**). Independent conductor-verification caught + fixed real defects in the agents’ output (a dashboard query bug; two security-suite defects — 1s-TTL provisioning + bare-wait hang).
- **Parallel round 2 (operator-directed, 5 background streams + conductor main-stream, all landed + pushed a58f7f8):**  **integration-port collision FIXED** (the round-1 follow-up) — cross-process flock (pg_itest_lock_test.go in store+rollout) serializes the shared Postgres so `go test -tags integration ./...` runs parallel-clean (§11.4.115 RED→GREEN, verified ×2);  **deviceemu 76→94%** (21 real error/orchestration tests);  **ota-protocol 99.3→100%** (submodule eda12b7);  **ota-artifact-validator 98.5→100%** (submodule 087fa08, §G2 security invariants);  **e2e tests/e2e/rollout_halt_safety.sh** (47/0/0 — the safety-critical rollout halt-on-breach path, previously happy-path-only);  **HelixQA bank → 11 challenges** (extended security wired, dry-run 11/0/0 + self-test).
- **REMAINING follow-ups (genuine, actionable, NOT hardware-blocked — DIMINISHING-RETURNS, do only if directed):** (a) coverage deepening on

the other owned submodules (ota-telemetry-schema, ota-rollout-engine) per §11.4.28; (b) store non-integration coverage (~45% — the pgx paths are integration-tagged, so this is largely a tag artifact; with -tags integration store is 88.7%); (c) additional e2e scenarios (delta-apply, multi-device rollout). NOTE: core/api/fabric/transport/deviceemu + both top bricks are now at high/100% coverage — further coverage work risks low-value filler (§11.4 anti-bluff), so prefer real feature/defect work or hardware-tier unblock over coverage-chasing. NOTE (§11.4.6 fact-correction): weasyprint 66.0 + pandoc 3.9 ARE present on this host — the prior “no weasyprint/LaTeX → PDF gap” note is stale. PDF siblings ARE generated via the documented §11.4.65 pipeline `pandoc <f>.md -s --metadata title=<base> -o <f>.html && weasyprint <f>.html <f>.pdf` (NOTE: scripts/export_docs.sh’s OWN PDF path needs a LaTeX engine, which IS genuinely absent — use the weasyprint pipeline for PDF).

- **Wave 6 shipped this session (2026-06-10, on main, all 4 upstreams):**
 - 16b4b23 android gitlink → ota-android-agent 26bd0a2 (:android AAR now builds — root-caused to an unpinned kotlin.android plugin, NOT MPP; :core 47/0 green).
 - 12087bc emulator failure→recall(forward-fix)→recovery e2e (TestRecallRecoveryE2E PASS +-race; podman shell variant tests/emulator/tier1_recall_recovery_e2e.sh syntax-clean, **conductor must run it live**).
 - 19e1346 dashboard §11.4.135 regression guard for the DeviceStatus.health-object crash (Vitest 58, RED/GREEN polarity).
 - 3c85b14 security probes recall+telemetry 28/0 (surfaced a STRONGER trust boundary — strict bindJSON rejects injected keys 400).
 - 88bd2c2 recall lifecycle e2e live 35/0 (tests/e2e/recall_lifecycle.sh).
 - 5123e67 HelixQA bluff audit + real bank runner (tools/helixqa/run_bank.sh): VERDICT HelixQA is NOT a bluff — canonical engine gates on dispatch-exit-0 + non-empty evidence ledger + catches its own negation; the real gap was an INCORPORATION gap (§11.4.27, never wired as a submodule / bank never machine-executed). Runner: dry-run 10/0/0 + self-test PASS. Audit: docs/research/helixqa_bluff_audit/REPORT.md.
 - 71be1cd per-event telemetry duration_ms+bytes_transferred end-to-end (closed the parked WIDEN row-4): ota-protocol gitlink → 3d360ab (additive, validation rejects negative), emulator emits → server ingest → store (memory+pgx) → read view; TestEmulatorTelemetryFieldsEndToEnd exact-value PASS + real-Postgres parity (TestPostgresRepositoryContract_Integration, podman PG, 0 skips).
 - 8a03a63+de87e21 emulation test-fabric research + full design (docs/research/emulation_infra/REPORT.md cited; docs/design/emulation_fabric/{DESIGN,ROADMAP,TEST_COVERAGE_PLAN}.md+SCHEMA.sql) — tiers T0–T3/Tfw/Tcp, extends containers submodule, honest macOS-M3 KVM gaps.
- **Queue done this session (all pushed, all 4 upstreams):** ✓ HelixQA LIVE full-bank 10/0 (d5fcf4a — fixed run_bank LIVE-mode <pw> substitution + free-port isolation); ✓ podman recall→recovery e2e 7/7 (71c3f77); ✓ HelixQA wired as a submodule at submodules/helixqa pinned latest fe17c08, gate Part C PASS, install_upstreams done (d488e6e, §11.4.27); ✓ emulation P1 T1-tier AVD-HVF boot smoke arm64-v8a booted+accel-proven (5d6b3ae).

- **Wave 7 (5-agent parallel, all pushed):** ✓ **fabric target-registry** project-side Go (c709c92 — server/internal/fabric + store memory+pgx, exclusive-lease §11.4.119 + non-empty-evidence §11.4.69, real-PG integration 0 skips, §1.1 mutation); ✓ **on-device AVD instrumentation test** → ota-android-agent 1061015 (8266355 gitlink; OK 5 tests 3x; UpdateEngine-present-but-degrades-to-Failed FACT); ✓ **P2 Tfw QEMU smoke** authored, honest SKIP (qemu absent — brew install qemu to activate) (504c92b); ✓ **helix-deps.yaml §11.4.31** + native test_cases bank (607e7c3); ✓ spec_impl_alignment row-4 RESOLVED + coverage_ledger Rev2 (7a7cb48).
- **Overnight stability hardening (2026-06-10, all pushed):** ✓ **P2 QEMU firmware tier ACTIVATED** — brew install qemu (11.0.1), tier_fw_qemu_smoke.sh real edk2 UEFI boot PASS (93ac0a3); ✓ **run_bank --self-test wired into the pre-build gate** (60da61a, §1.1 every build); ✓ **Challenges submodule incorporated** at submodules/challenges (438057d, §11.4.27, gate PASS, helix-deps 11 deps); ✓ **comprehensive stability sweep ALL GREEN** on HEAD 438057d (Go all-tiers+race, pgx integration, meta-test, dashboard 58, HelixQA LIVE bank 10/0, podman full-lifecycle + fleet 5/5 + recall-recovery, QEMU tier, AVD boot + on-device, submodule cores) — see docs/qa/STABILITY_REPORT.md. DEFERRED zero-risk: fabric scheduler P3 (would be unwired §11.4.124), HelixQA in-tree compile (./containers layout, HelixQA-side). [CORRECTED 2026-06-10 overnight: the prior “PDF doc siblings (no weasyprint/LaTeX)” deferral was stale — weasyprint 66.0 IS present; PDF siblings for the handoff docs ARE now generated via pandoc+weasyprint; only a full-corpus PDF backfill remains as optional follow-up.]
- **Remaining NEXT (real PWUs):** (1) emulation **P4 Cuttlefish** (Linux+KVM) for real update_engine A/B + the **GSI-A/B real-apply** question (REPORT §2.2/§8 — on a google_apis AVD UpdateEngine is present but unusable for a non-system caller); (2) **P2 QEMU activate** (brew install qemu → tier_fw_qemu_smoke.sh goes green); (3) **P3 fabric scheduler** wiring on top of the registry (Nomad/LAVA backend); (4) wire HelixQA’s missing challenges submodule (its go.mod replace digital.vasic.challenges points at an absent submodules/challenges); (5) HelixQA: run_bank.sh --self-test into the pre-build gate, Dispatcher→autonomous-runner (§11.4.124); (6) OPTIONAL full-corpus PDF backfill: weasyprint 66.0 + pandoc 3.9 ARE present, and the handoff docs’ .pdf siblings are now generated via pandoc -s ... && weasyprint; a one-shot pass to (re)generate .pdf siblings across the ENTIRE docs corpus (not just handoff docs) is the only remaining §11.4.65 PDF item — low-priority, non-blocking.
- **Wave 5 shipped (on main):** 2391cb6 **full OTA lifecycle + multi-device fleet PROVEN on podman** (single device 1.0.0→1.1.0 with the complete download→success telemetry accepted; fleet 5/5 → 1.0.0→1.2.0; evidence docs/qa/20260610T11191*/ + ...111928Z-fleet/); f5a3428 N-device scaling (50 concurrent, ~1580 ops/sec, 0 err, -race); 4658125 dashboard Fleet-detail + a real product bug fix (DeviceStatus.health object). **The emulator now fully runs+tests the codebase end-to-end on podman — the prior “honest gap” is closed.**
- **NEXT non-blocked candidates:** richer telemetry ingest fields (duration_ms/ bytes_transferred — needs the device to send them); a rollback/recall lifecycle e2e on the emulator (drive a failure event → recall forward-fix); migrate the dashboard health-bug to a Fixed.md ATM entry. **BLOCKED:** ATM-003 Tier-2 (Linux+KVM Cuttlefish, design ready in docs/design/CUTTLEFISH_TIER2.md) + ATM-004 Tier-3 (RK3588 hardware).

- **Waves 3–4 also shipped (on main):** 3c57867 closed both protocol gaps (UpdateAvailable.deployment_id [ota-protocol→7920842] + GET / deployments); 8c0521d Tier-1 podman container e2e (PROVEN: control-plane container boots, ota-device-emu container runs the real round-trip — evidence docs/qa/20260610T105306Z/); 5d4920e dashboard ArtifactUpload + populated-detail (Vitest 50, Playwright 20); submodule pins bumped — constitution ba0f702, ota-rollout-engine 7a90912, ota-artifact-validator 77c6b48, containers 845ad45. Tracker live at docs/Issues.md (ATM-003/004 Operator-blocked: Tier-2 Cuttlefish-on-Linux-KVM / Tier-3 RK3588 hardware) + docs/Fixed.md.
- **Just-shipped this session (on main, pushed all 4 upstreams):**
 - 50ef5c6 — feat(api): per-device telemetry filters + group/members pagination (OpenAPI synced, redocly-clean).
 - b0b8ee2 — chore(dashboard): sync API client to new pagination/filter params (dashboard client lockstep).
 - 7dc3334 — feat(emulator): Tier-1 Go OTA device-emulator + resilience (server/internal/deviceemu + cmd/ota-device-emu; surfaced the telemetry deployment_id protocol gap).
 - fa571b8 — test(dashboard): comprehensive UI testing system (Vitest 43 + Playwright 17 + ally).
 - a839220 — test(qa): autonomous e2e (50/50) + security (39/39) + HelixQA bank.
 - **containers submodule** advanced to 845ad45 — feat(emulator): ota-device-emu runtime image + on-demand boot recipe (pushed to its github origin); parent gitlink bumped in this commit.
- **Branch:** main. All work merges to main; commit + push fan out to all upstreams.
- **Upstreams (4):** github (git@github.com:HelixDevelopment/helix_ota.git, also origin fetch), gitlab (git@gitlab.com:helixdevelopment1/helix_ota.git), gitflic (git@gitflic.ru:helixdevelopment/helix_ota.git), gitverse (git@gitverse.ru:helixdevelopment/helix_ota.git). origin push fans out to all four.
- **No in-flight background PIDs / detached pushes known at handoff time** — verify with git status + check qa-results/push_failures/ per §11.4.88 before assuming clean.
- **Container runtime on this host:** podman (the pgx integration + dev stack boot via the containers submodule on podman). **No host QEMU / no AVD / no nested KVM on this Apple-Silicon host** — see binding constraints.

2. Current PHASE + immediate NEXT + terminal goal

- **PHASE (ACTIVE 2026-06-11): RK3588 / Orange Pi 5 Max A/B device emulator** — operator mandate “implement everything you can for Linux and if possible for macOS; we will provide a Linux host.” Research DONE (docs/research/rk3588_emulator/REPORT.md, cited §11.4.123): the dev-host-feasible path on THIS Mac is a QEMU virt+HVF Linux guest + **U-Boot bootcount/altbootcmd auto-rollback + RAUC format=verity dm-verity A/B slots** (reuse RAUC+U-Boot §11.4.74; extend containers/pkg/vm). Real Android update_engine A/B stays Linux+KVM-gated (Cuttlefish T2 — for the incoming Linux host).

- **Immediate NEXT (priority order — RK3588 emulator PWUs per REPORT §6):**

1. **macOS dev-host A/B-virt tier:** tests/emulator/ab_virt/build_image.sh builds the base aarch64 Buildroot guest in a podman container (macOS-portable: named container + named-volume DL cache + podman cp extract — NO /Volumes/T7 bind-mount, NO :Z; output gitignored §11.4.77). **STATE (HEAD 0d27438):**  PWU-AB-0 base aarch64 Buildroot image BUILT;  PWU-AB-1 FOUNDATION GREEN — tests/emulator/ab_virt/boot_smoke.sh boots it under qemu-system-aarch64 -M virt -accel hvf to a LIVE interactive userspace (kernel on real Apple CPU MIDR 0x610f + getty login + post-login uname + interactive sentinel; 196-line evidence docs/qa/20260611T061626Z-ab-virt-boot/console.log, anti-bluff §11.4.107);  A/B slot mechanism AUTHORED (parse-clean + coherent, NOT yet run): tests/emulator/ab_virt/assemble_ab_disk.sh (2-slot GPT builder, stamps out/.disk_ok) + tests/emulator/ab_virt/u-boot_ab/boot.cmd (BOOT_ORDER/bootcount/altbootcmd slot-select);  4 parallel streams committed — HelixQA bank HOTA-AB-VIRT-B00T (dry-run 12/0/0), §11.4.45 Status docs docs/emulator/rk3588_ab_virt/, Cuttlefish PWU-CF-2 corrupt-slot rollback, the A/B-authoring. ⌚ build8 (build_image.sh + U-Boot qemu_arm64 + RAUC + libssl-dev/bison/flex) building u-boot.bin. Root-caused+fixed en route: orphaned-rootless-podman-storage disk overflow (28GB), invalid Bootlin toolchain, single-quote/paren shell-syntax bug, U-Boot openssl/evp.h missing-libssl-dev. IMMINENT NEXT (HEADLINE, gated only on build8's u-boot.bin): run assemble_ab_disk.sh → boot via U-Boot (-bios u-boot.bin -drive ab_disk.img) → prove root==slot-B after apply+flip+reboot (RED→GREEN §11.4.115, captured evidence). Then PWU-AB-2 (RAUC dm-verity), PWU-AB-3 (corrupt-slot auto-rollback — headline), PWU-AB-4 (agent ApplyPort over ota-protocol), PWU-AB-5 (chaos + HelixQA bank). Build hygiene: named container + podman rm -f helix_ab_build after each build; NO apostrophes/parens inside the single-quoted podman bash -c; U-Boot needs libssl-dev/bison/flex in the build container.
 2. **Linux/Cuttlefish T2 tier:** implement the /dev/kvm-gated harness (§11.4.3/§11.4.81) from docs/design/CUTTLEFISH_TIER2.md — honest SKIP here, ready to verify on the incoming Linux host (real update_engine A/B + AVB/dm-verity + auto-rollback).
 3. Older NEXT-wave (hardware/ingest-gated): device-side TUF; row-4 richer telemetry fields. NOTE: this is a multi-DAY effort (research est. ~5-9 eng-days, mostly the iterative Buildroot guest image) — proceeds across loop cycles; report “all done” only when genuinely proven (§11.4.6).
- **Terminal goal (loop stop condition A, §11.4.126):** a new fully-validated-and-verified version (tag) created AND published across all owned submodules + main repo to all 4 upstreams.

3. Binding constraints (do not violate)

- **Anti-bluff covenant §11.4** — every PASS carries positive captured evidence; metadata-only / config-only / absence-of-error / grep-without-runtime PASS forbidden. Tests AND HelixQA Challenges bound equally.
- **Absolute no-force-push §11.4.113** — force-push (--force, --force-with-lease, +<ref>) is STRICTLY forbidden on every repo/submodule. Integrate via fetch → merge-onto-latest-main → fast-forward push.

- **podman-only on this host** — use the containers submodule (vasic-digital/containers, §11.4.76) for any containerized workload; never host-direct emulator/adb/qemu (§11.4.109 guard).
- **Tier-2 Android A/B is host-gated** — real update_engine A/B + AVB/dm-verity auto-rollback needs Cuttlefish on a Linux + nested-KVM box; the Apple-Silicon applehv host cannot run it. NOT structurally impossible (§11.4.112) — host/hardware-gated only.
- **Exact naming/versions:** Go module `github.com/HelixDevelopment/helix_ota/server`; submodules under `submodules/`; constitution submodule at `constitution/`. Toolchain: Go 1.26.2, Gradle 9.5 + Kotlin 2.3.20 (AGP 8.5.2 + plain `kotlin.android` builds; Kotlin MPP does NOT). No LaTeX/drawio/plantuml/graphviz/docker on host (so `export_docs.sh`'s LaTeX-based PDF path is unavailable) — BUT PDF siblings ARE produced via `pandoc -s + weasyprint 66.0` (both present); use that pipeline for §11.4.65 PDF.
- **Commit/push discipline §2.1 + §11.4.88** — commit + push to ALL 4 upstreams; pushes may run detached.

How a fresh session resumes from this file alone

Given only this file's path, a new agent: reads §0 handoff docs, runs `git fetch --all`, confirms HEAD against §1, picks up the §2 PHASE/NEXT, and works under the §3 constraints toward the terminal goal — zero additional context required.